

D600

Statistical Data Mining

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A telecom company with 4 million subscribers has customer records with no outcome labels and wants to discover natural customer segments for marketing. Its data warehouse was migrated last year. Which approach fits?

- A. Logistic regression
- B. Unsupervised clustering, such as k-means
- C. Linear regression on monthly revenue
- D. Supervised learning, since all data mining needs a target

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A lender with branches in 11 states wants to predict whether each applicant will default (yes or no) using income, debt ratio, and credit history. The model will be reviewed by the risk committee each quarter. Which task and model type fit?

- A. Regression, using linear regression to predict default
- B. Clustering applicants into groups
- C. Classification, using logistic regression
- D. Time-series forecasting, since reviews happen quarterly

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A model's output is a predicted monthly energy bill in dollars. What kind of supervised task is it?

- A. Association rule mining
- B. Clustering
- C. Regression
- D. Classification

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A property-tax consultancy with 30 appraisers uses the model $\text{price} = 50,000 + 120 \times (\text{square feet}) + 8,000 \times (\text{bedrooms})$. What is the predicted price for a 1,500-square-foot, 3-bedroom home?

- A. \$254,000
- B. \$206,000
- C. \$230,000
- D. \$204,000

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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5. [TRANSFER — reversed stem / different variable]

Using $\text{price} = 50,000 + 120 \times (\text{square feet}) + 8,000 \times (\text{bedrooms})$, a 3-bedroom home is predicted at \$278,000. What square footage does the model imply?

- A. 1,900 square feet
- B. 2,100 square feet
- C. About 2,317 square feet
- D. 1,700 square feet

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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6.

An analyst at a 5-store furniture chain adds 12 randomly generated noise variables to a sales regression, and R^2 rises from 0.71 to 0.74. The chain's peak season is spring. What is the best interpretation?

- A. Refit the model on spring data only.
- B. The model is better, because a higher R^2 always means a better model.
- C. Adjusted R^2 must have risen even more than R^2 .
- D. The rise is expected even from noise; check adjusted R^2 , which likely fell.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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7.

A fare model at a regional airline includes both flight distance in miles and flight distance in kilometers. The airline operates 60 aircraft. The two distance coefficients swing wildly between samples and have very large standard errors. What is the diagnosis?

- A. Too few aircraft in the data.
- B. Multicollinearity; drop one of the distance variables.
- C. Heteroscedasticity; transform the fares.
- D. Underfitting; add more predictors.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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8.

A churn logistic regression at a gym chain with 38 locations has a coefficient of 0.693 on the flag 'missed 3+ weeks.' The monthly fee is \$49. What does the coefficient imply?

- A. Members who missed 3+ weeks have about double the odds of churning.
- B. The odds of churn rise by 69.3%.
- C. The probability of churn rises by 0.693.
- D. Churn rises by \$49 per member.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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9. [TRANSFER — reversed stem / different variable]

A logistic regression coefficient is -0.693 . What happens to the odds of the outcome for a one-unit increase in that variable?

- A. The odds are cut roughly in half.
- B. The odds fall by 69.3%.
- C. The odds double.
- D. The outcome becomes impossible.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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10.

An analyst at a 12-person analytics shop plans to run PCA on customer data where income ranges from \$0 to \$250,000 and store visits per month range from 0 to 30. The data was collected by an outside vendor. What should be done first?

- A. Run PCA on the raw values.
- B. Re-collect the data in-house.
- C. Standardize each variable (mean 0, standard deviation 1).
- D. Remove the visits variable because its range is small.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

PCA returns explained variance of 45%, 25%, 15%, 10%, and 5% for components 1–5. How many components are needed to keep at least 80% of the variance?

- A. 5, because PCA always keeps every component
- B. 3
- C. 2
- D. 4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A demand model at a snack maker, built on three years of data from 200 stores, scores $R^2 = 0.98$ on training data and 0.52 on held-out test data. The team lead recently joined from a consulting firm. What is most likely?

- A. Trust the training score, since it uses more data.
- B. Underfitting; add more variables.
- C. The consulting background explains the gap.
- D. Overfitting; simplify or regularize and use cross-validation.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A telecom company ~~with 4 million subscribers~~ has customer records with no outcome labels and wants to discover natural customer segments for marketing. ~~Its data warehouse was migrated last year.~~ Which approach fits?

Target word: supervised or unsupervised **Decides:** No labels; the goal is to find groups.

A	Adjacent category	Needs a labeled yes/no outcome.
B	Correct	Finds segments without a target variable.
C	Confusion set	Predicts a number; it does not discover segments.
D	Absolute language	Unsupervised methods exist for exactly this case.

2. Answer C Concept 2

A lender ~~with branches in 11 states~~ wants to predict whether each applicant will default (yes or no) using income, debt ratio, and credit history. ~~The model will be reviewed by the risk committee each quarter.~~ Which task and model type fit?

Target word: task type **Decides:** The target is binary (default yes/no).

A	Confusion set	Linear regression predicts continuous values and can go outside 0-1.
B	Adjacent category	Unsupervised; ignores the known default labels.
C	Correct	A yes/no target is a classification problem.
D	Camouflage	The review schedule does not define the prediction task.

3. Answer C Concept 2 · transfer

A model's output is a predicted monthly energy bill in dollars. What kind of supervised task is it?

Target word: task type from the output **Decides:** The output is a continuous number.

A	Confusion set	Finds items that occur together, not a numeric prediction.
B	Adjacent category	Unsupervised and produces groups, not amounts.
C	Correct	Predicting a numeric amount is regression.
D	Inverted operation	Classification predicts categories.

4. Answer A Concept 3

A property-tax consultancy ~~with 30 appraisers~~ uses the model $\text{price} = 50,000 + 120 \times (\text{square feet}) + 8,000 \times (\text{bedrooms})$. What is the predicted price for a 1,500-square-foot, 3-bedroom home?

Target word: prediction from the equation **Decides:** Plug all terms in, including the intercept.

A	Correct	$50,000 + 120 \times 1,500 + 8,000 \times 3 = 50,000 + 180,000 + 24,000$.
B	Inverted operation	Subtracts the bedroom term: $50,000 + 180,000 - 24,000$.
C	Dropped element	Leaves out the bedroom term.
D	Dropped element	Leaves out the \$50,000 intercept.

5. Answer D Concept 3 · transfer

Using $\text{price} = 50,000 + 120 \times (\text{square feet}) + 8,000 \times (\text{bedrooms})$, a 3-bedroom home is predicted at \$278,000. What square footage does the model imply?

Target word: solve for an input **Decides:** Remove intercept and bedroom term, then divide by 120.

A	Dropped element	Forgets the bedroom term: $228,000 / 120$.
B	Inverted operation	Adds the bedroom term instead of subtracting it: $252,000 / 120$.
C	Dropped element	Divides the full price by 120, ignoring intercept and bedrooms.
D	Correct	$(278,000 - 50,000 - 24,000) / 120 = 204,000 / 120 = 1,700$.

6. Answer D Concept 4

An analyst at a 5-store furniture chain adds 12 randomly generated noise variables to a sales regression, and R^2 rises from 0.71 to 0.74. ~~The chain's peak season is spring.~~ What is the best interpretation?

Target word: meaning of the R^2 rise **Decides:** R^2 cannot fall when variables are added.

A	Camouflage	Seasonality does not explain an R^2 gain from noise.
B	Absolute language	R^2 rises mechanically with added variables.
C	Inverted operation	Adjusted R^2 is never above R^2 and drops for useless variables.
D	Correct	Adjusted R^2 penalizes useless variables.

7. Answer B Concept 5

A fare model at a regional airline includes both flight distance in miles and flight distance in kilometers. ~~The airline operates 60 aircraft.~~ The two distance coefficients swing wildly between samples and have very large standard errors. What is the diagnosis?

Target word: diagnosis **Decides:** Two predictors carry the same information.

A	Camouflage	Fleet size is not what makes the coefficients unstable.
B	Correct	Perfectly correlated predictors make coefficients unstable.
C	Confusion set	Unequal error variance is a different problem.
D	Adjacent category	Adding variables would not fix redundant ones.

8. Answer A Concept 6

A churn logistic regression at a gym chain ~~with 38 locations~~ has a coefficient of 0.693 on the flag 'missed 3+ weeks.' ~~The monthly fee is \$49.~~ What does the coefficient imply?

Target word: effect on odds **Decides:** $e^{0.693} \approx 2$.

A	Correct	Exponentiating the coefficient gives an odds ratio of about 2.
B	Dropped element	Skips exponentiation.
C	Confusion set	Coefficients are in log-odds, not probability.
D	Camouflage	The fee does not enter the coefficient's meaning.

9. Answer A Concept 6 · transfer

A logistic regression coefficient is -0.693 . What happens to the odds of the outcome for a one-unit increase in that variable?

Target word: negative coefficient to odds **Decides:** $e^{-0.693} \approx 0.5$.

A	Correct	An odds ratio of 0.5 halves the odds.
B	Dropped element	Skips exponentiation.
C	Inverted operation	That is the effect of $+0.693$.
D	Absolute language	Odds shrink but stay above zero.

10. Answer C Concept 7

An analyst ~~at a 12-person analytics shop~~ plans to run PCA on customer data where income ranges from \$0 to \$250,000 and store visits per month range from 0 to 30. ~~The data was collected by an outside vendor.~~ What should be done first?

Target word: preprocessing for PCA **Decides:** Variables are on very different scales.

A	Dropped element	Income would dominate simply because of its units.
B	Camouflage	The source of the data does not decide the preprocessing step.
C	Correct	Otherwise income's large variance dominates the components.
D	Inverted operation	Small units do not mean small importance; scaling fixes this.

11. Answer B Concept 7 · transfer

PCA returns explained variance of 45%, 25%, 15%, 10%, and 5% for components 1-5. How many components are needed to keep at least 80% of the variance?

Target word: count components by cumulative variance **Decides:** $45 + 25 = 70$; $+ 15 = 85$.

A	Absolute language	The point of PCA is to keep fewer components.
B	Correct	Cumulative variance first reaches 80% (85%) at component 3.
C	Dropped element	Two components reach only 70%.
D	Adjacent category	Counts every component at 10% or more rather than the cumulative target.

12. Answer D Concept 8

A demand model at a snack maker, ~~built on three years of data from 200 stores,~~ scores $R^2 = 0.98$ on training data and 0.52 on held-out test data. ~~The team lead recently joined from a consulting firm.~~ What is most likely?

Target word: diagnosis from train vs. test **Decides:** Large gap between training and test performance.

A	Dropped element	Held-out performance is what predicts real use.
B	Inverted operation	Underfitting shows poor training performance too.
C	Camouflage	Who leads the team does not diagnose the model.
D	Correct	The model memorized training data and does not generalize.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	9	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Supervised vs. unsupervised	Supervised learning needs a labeled target to predict; unsupervised learning finds structure (clusters, components) without labels.	Reaching for a predictive model when there is no target to learn from.	Whether the data has a known outcome column.
2	Classification vs. regression	A categorical target calls for classification (such as logistic regression); a numeric target calls for regression (such as linear regression).	Using linear regression for a yes/no outcome or calling a dollar prediction classification.	The type of the outcome: yes/no, category, or amount.
3	Using a linear regression equation	Prediction = intercept + $\Sigma(\text{coefficient} \times \text{value})$; each coefficient is the change in the prediction per unit, holding other variables constant.	Dropping the intercept or a term, or subtracting a positive term.	A fitted equation and a set of input values.
4	R² vs. adjusted R²	R ² never falls when variables are added; adjusted R ² penalizes variables that add little and can fall.	Treating a higher R ² from more variables as proof of a better model.	Variables added and R ² rising slightly.
5	Multicollinearity	Highly correlated predictors inflate standard errors and make coefficients unstable; remove or combine them (check VIF).	Mistaking unstable coefficients for overfitting or heteroscedasticity.	Two predictors that measure nearly the same thing; coefficients that swing between samples.
6	Logistic coefficients and odds	A logistic coefficient b changes the odds by a factor of e^b per unit: $e^{0.693} \approx 2$ (double), $e^{-0.693} \approx 0.5$ (half).	Reading b as a change in probability or as a percentage without exponentiating.	A logistic coefficient near ± 0.693 , or a question about odds.
7	PCA and dimension reduction	Standardize variables before PCA; components are ordered by variance explained, and you keep enough to reach the target cumulative share.	Running PCA on raw scales so one large-unit variable dominates, or counting components wrong.	Variables on very different scales, or a list of variance percentages.
8	Overfitting and holdout testing	A large gap between training and test performance signals overfitting; judge models on held-out data and simplify or regularize.	Trusting the training score or calling the gap underfitting.	Training score far above test score.

Course scope (public description): Statistical Data Mining focuses on concepts in data preparation and supervised and unsupervised machine learning techniques. The course helps students gain basic knowledge in statistics, data preparation, regression, and dimensional reduction. Learners implement supervised models—specifically classification and prediction data mining models—to identify relationships among variables that are not apparent with more surface-level techniques. The course also explains when, how, and why to use unsupervised models to best meet organizational needs. The following course is a prerequisite: Data Preparation and Exploration.